

2. Precise typing of pattern matching (II)

Boolean type connectives are needed to type pattern matching:

`match e with p1 -> e1 | p2 -> e2`

Suppose that $e : T$ and let us write $T_1 \searrow T_2$ for $T_1 \wedge \neg T_2$

- To infer the type T_1 of e_1 we need $T \wedge \{p_1\}$;
- To infer the type T_2 of e_2 we need $(T \searrow \{p_1\}) \wedge \{p_2\}$;
- The type of the match expression is $T_1 \vee T_2$.
- Pattern matching is exhaustive if $T \leq \{p_1\} \vee \{p_2\}$;

Formally:

$$\begin{array}{c}
 \text{[Match]} \\
 \frac{\Gamma \vdash e : T \quad \Gamma, T \wedge \{p_1\} / p_1 \vdash e_1 : T_1 \quad \Gamma, T \searrow \{p_1\} / p_2 \vdash e_2 : T_2}{\Gamma \vdash \text{match } e \text{ with } p_1 \rightarrow e_1 \mid p_1 \rightarrow e_2 : T_1 \vee T_2} \quad (T \leq \{p_1\} \vee \{p_2\})
 \end{array}$$

where T/p is the type for the capture variables in p when it is matched against values in T .